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MATH 511

Need: basic linear algebra
some experience with inner product spaces
(220/420)
Have an undergraduate book handy..

Textbook "Bible" of matrix theory.

Read Familiarize yourselves with book.
Read Chapter 0

Lectures Self-contained.
Scan & post notes
Recorded, zoom available. (Canvas)

Material

Chapter 1: eigenvalues, similarity
diagonalization.

Chapter 2: Unitary equivalence, normal matrices
Schur's Theorem applications.

Chapter 3: Jordan Canonical Form

Chapter 4: Hermitian matrices

Chapter 5: Norms

Chapter 6: Location & perturbation
of eigenvalues

Chapter 1

Read 1.0–1.2 in detail.

§1.1 Summarize notation/conventions.

- $M_n(\mathbb{F})$: all $n \times n$ matrices with entries from field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$)
  reals, complex numbers
- $M_n := M_n(\mathbb{C})$
- $A = [a_{ij}] \in M_n$
- A^T transpose of A : rows $\longrightarrow$ columns

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

- A^* conjugate transpose of A
  (or hermitian adjoint)

$$A = [a_{ij}] \in M_n$$

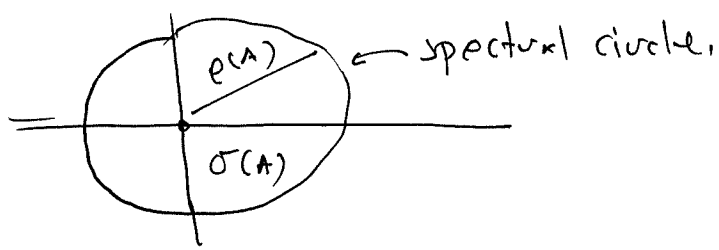
$$A^* = \overline{(A^T)} = [\bar{a}_{ji}]$$

$$\begin{bmatrix} 1-i & i \\ 2 & 3i \end{bmatrix}^* = \begin{bmatrix} \overline{1-i} & \bar{2} \\ \bar{i} & \overline{3i} \end{bmatrix}$$

$$= \begin{bmatrix} 1+i & 2 \\ -i & -3i \end{bmatrix}$$

- $\sigma(A)$ = <u>spectrum</u> of $A \in M_n$
  = set of all eigenvalues of A
  $= \{ \lambda \in \mathbb{C} : Ax = \lambda x$ for some $x \in \mathbb{C}^n \setminus \{0\} \}$

- $\rho(A)$ = <u>spectral radius of $A \in M_n$</u>
  (rho of A)
  $= \max\{ |\lambda| : \lambda \in \sigma(A) \}$



- Recall: $A \in M_n$. Then

$$Ax = \lambda x \iff (A - \lambda I)x = 0$$
$$\iff x \in N(A - \lambda I)$$

nullspace.

If $x \neq 0$ & $x \in N(A - \lambda I)$, then x is an eigenvector of A corresponding to $\lambda \in \mathbb{C}$.

We then call
$N(A - \lambda I)$ = eigenspace of λ.

- Given a scalar polynomial

$$p(t) = a_k t^k + a_{k-1} t^{k-1} + \cdots + a_1 t + a_0$$

$$a_0, a_1, \ldots, a_k \in \mathbb{C}$$

& given $A \in M_n$, by "$p(A)$" we mean

$$p(A) = a_k A^k + a_{k-1} A^{k-1} + \cdots + a_1 A + a_0 I \in M_n$$

<u>Fact</u> If $Ax = \lambda x$, $x \neq 0$ (i.e. $\lambda \in \sigma(A)$) then

$$p(A)x = a_k A^k x + a_{k-1} A^{k-1} x + \cdots + a_1 A x + a_0 x$$

$$= a_k \lambda^k x + a_{k-1} \lambda^{k-1} x + \cdots + a_1 \lambda x + a_0 x$$

$$= p(\lambda) x$$

i.e. $\boxed{\lambda \in \sigma(A) \implies p(\lambda) \in \sigma(p(A))}$

- Conditions equivalent to $A \in M_n$ being singular (i.e. non-invertible)

(nonsingular $\iff$ invertible)

A singular $\iff \det A = 0$

$\iff$ columns are lin. dependent

$\iff A^{-1}$ does not exist

$\iff N(A) \neq \{0\}$

i.e. $\exists\, x \neq 0 : Ax = 0$

$\iff \lambda = 0 \in \sigma(A) \quad (Ax = 0 \cdot x,\ x \neq 0)$

$\iff$ Every consistent system $Ax = b$ has an infinite number of solutions.

§ 1.2 The characteristic polynomial

$A \in M_n$. To find eigenvalues we "solve"

$$\boxed{\det(tI - A) = 0}$$

for t. This is the characteristic equation / $\det(tI-A)$ characteristic polynomial. The roots are

$$\lambda_1, \lambda_2, \ldots, \lambda_n$$

which form the spectrum $\sigma(A)$

($\det(tI-A)$ = is n^{th} degree monic polynomial in t)

(monic = coefficient of t^n is 1)

Notes

1) $A \in M_n(\mathbb{R})$, the $\sigma(A)$ is conjugate closed.

2) $\det(A - tI) = \det(-(tI-A))$

$$= (-1)^n \det(tI - A)$$

Knowing the eigenvalues we solve

$$(A - \lambda_j I)\, x_j = 0$$

to find eigenvector(s) associated to λ_j $(j = 1, 2, \ldots, n)$

<u>Note</u> For $n \geq 5$ we can only approximate the eigenvalues.

$$\boxed{\text{Important connections exist, between entries \& eigenvalues.}}$$

We need two concepts:

(1) k-th <u>elementary</u> <u>symmetric</u> <u>function</u> of $\lambda_1, \lambda_2, \ldots, \lambda_n$

$$S_k = S_k(\lambda_1, \lambda_2, \ldots, \lambda_n) = \sum_{\substack{1 \leq i_1 < \cdots < i_k \leq n \\ \uparrow \\ \text{all } k\text{-tuples}}} \prod_{j=1}^{k} \lambda_{i_j}$$

Example: $k=4$, $n=6$

choose $i_1, i_2, i_3, i_k \in \{1,2,3,\ldots,6\}$

e.g. $(1, 3, 5, 6)$

$$\prod_{j=1}^{k} \lambda_{i_j} = \lambda_{i_1} \lambda_{i_2} \lambda_{i_4} \lambda_{i_4}$$

$$= \lambda_1 \lambda_3 \lambda_5 \lambda_6$$

Eg $n=3$

$$S_1 = \lambda_1 + \lambda_2 + \lambda_3$$

$$S_2 = \lambda_1 \lambda_2 + \lambda_1 \lambda_3 + \lambda_2 \lambda_3$$

$$S_3 = \lambda_1 \lambda_2 \lambda_3$$

(2) Principal Submatrices & Principal minors of a matrix

Let $A = [a_{ij}] \in M_n$

Let $\alpha \subseteq \{1, 2, \ldots, n\}$ with elements in increasing order.

$\beta \subseteq \{1, 2, \ldots, n\}$ also with elements in increasing order.

By:

- $A[\alpha|\beta]$ = <u>submatrix of A</u> obtained by keeping rows of A indexed by α & columns of A indexed by β

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad \alpha = \{1, 3\}, \quad \beta = \{2, 3\}$$

$$A[\alpha|\beta] = \begin{bmatrix} 2 & 3 \\ 8 & 9 \end{bmatrix}$$

$$A[\{1\}|\{1,2,3\}] = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$A[\{2\}|\{2\}] = 5$$

- If $\alpha = \beta$, then

$$A[\alpha|\beta] = A[\alpha|\alpha] =: A[\alpha]$$

is called a <u>principal</u> <u>submatrix</u> of A

(necessarily square)

- $\det A[\alpha]$ = <u>principal</u> <u>minor</u> of A

<u>Example</u>

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 0 \\ 10 & 11 & 12 & -1 \end{bmatrix}$$

$$\det A[\{1,4\}|\{1,4\}] = \det A[\{1,4\}]$$

$$= \det \begin{bmatrix} 1 & 0 \\ 10 & -1 \end{bmatrix} = -1 + 0 \cdot 10 = -1$$

$$\det A[\{3\}] = \det[9] = 9$$

$$\det A[\{1,2,3,4\}] = \det A = \cdots$$

Back to characteristic polynomial & eigenvalues.

Given $A \in M_n$, the sum of all principal minors of order k (i.e. determinants of $k \times k$ principal submatrices), $k = 1, 2, \ldots, n$ is denoted by

$$E_k(A) = \sum_{\substack{\alpha \subseteq \{1,2,\ldots,n\} \\ |\alpha| = k}} \det A[\alpha]$$

($|\alpha|$ = cardinality of α)

Fact 1 (Easy but laborious to prove – see algebraic facts)

If $p_A(t) = (t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_n)$ is the characteristic polynomial of $A \in M_n$,

then

$$\boxed{p_A(t) = t^n - E_1(A)\,t^{n-1} + E_2(A)\,t^{n-2} - \cdots \pm E_n(A)}$$

Fact 2 Like above,

$$\boxed{P_A(t) = t^n - S_1 t^{n-1} + S_2 t^{n-2} - \cdots \pm S_n}$$

S_k : k-th elem. symm. function

$$\boxed{S_j = E_j(A) \quad j = 1, 2, \ldots, n}$$

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Simple consequences

(of formulas for charact. polyn.)

$A \in M_n$, $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$

Then

- $S_n = E_n(A)$, i.e. $\lambda_1 \lambda_2 \cdots \lambda_n = \det(A)$.
- $S_1 = E_1(A)$, i.e. $\lambda_1 + \lambda_2 + \dots + \lambda_n = a_{11} + a_{22} + \dots + a_{nn}$

(sum of diagonal entries of $A = [a_{ij}]$)

($S_1 = E_1(A) =$ <u>trace of A</u>

$= tr(A) = trace(A)$.)

<u>Example</u>

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\lambda_1 + \lambda_2 = a_{11} + a_{22} = 1 + 4 = 5$$

$$\lambda_1 \lambda_2 = \det(A) = 1 \cdot 4 - 2 \cdot 3 = -2$$

i.e. $P_A(t) = \det(tI - A) = t^2 - 5t - 2$

§ 1.3 Similarity

Definition $A, B \in M_n$ are called <u>similar</u> ($A \sim B$) if there exists invertible $S \in M_n$ such that

$$B = S^{-1} A S$$

(or equiv. $A = S B S^{-1}$)

(call: $A \longrightarrow S^{-1} A S$ a similarity transformation)

Clearly, similarity ($\sim$) is an "equivalence relation":

(a) reflexive $A \sim A$ $(S = I)$

(b) symmetric $A \sim B \iff B \sim A$

$(B = S^{-1} A S \iff A = S B S^{-1})$

(c) transitive

$$\left.\begin{array}{l} A \sim B \\ B \sim C \end{array}\right\} \Rightarrow A \sim C$$

$$\left.\begin{array}{l} A \sim B \Rightarrow B = S^{-1} A S \\ B \sim C \Rightarrow C = R^{-1} B R \end{array}\right\} \Rightarrow C = R^{-1} B R = R^{-1}(S^{-1} A S) R$$

$$= (SR)^{-1} A (SR)$$

$$\Rightarrow A \sim C.$$

The equivalence classes have a name in similarity: "orbits"

$$\text{Orbit of } A = \{ B \in M_n : A = S^{-1} B S,\ S \in M_n \text{ invertible} \}$$

Orbits are useful to know because similarity preserves a lot of properties:

<u>Theorem</u> If $A \sim B$, then

$$P_A(t) = P_B(t).$$

<u>Proof</u>

$$\begin{aligned}
P_B(t) &= \det(tI - B) \\
&= \det(tI - SAS^{-1}) \quad (B = SAS^{-1}) \\
&= \det(tSS^{-1} - SAS^{-1}) \\
&= \det(S[tI - A]S^{-1}) \\
&= \det S \cdot \det(tI - A) \cdot \det S^{-1} \\
&\quad \text{(property of determinants)} \\
&= \cancel{\det S} \cdot \det(tI - A) \cancel{\frac{1}{\det S}} \\
&= P_A(t) \quad \blacksquare
\end{aligned}$$

<u>Corollary</u> If $A \sim B$, then $\sigma(A) = \sigma(B)$

(similarity preserves eigenvalues)

(Thus it preserves $S_1, \dots, S_n$, $E_1(A), \dots, E_n(A)$)

<u>Facts</u>

(i) If $A \sim B$, then $\lambda \in \sigma(B) \iff \lambda \in \sigma(A)$,

moreover,

$$\lambda \in \sigma(B) \Rightarrow \exists x \neq 0 : Bx = \lambda x$$

$$\Rightarrow \exists x \neq 0 : SAS^{-1}x = \lambda x$$

$$\Rightarrow \exists x \neq 0 : A(S^{-1}x) = \lambda(S^{-1}x)$$

i.e. $\lambda \in \sigma(A)$ but <u>also</u> $S^{-1}x$ is the corresponding eigenvector.

Similarity transforms eigenvectors predictably!

$$\boxed{x \in N(B - \lambda I) \iff S^{-1}x \in N(A - \lambda I) \text{ when } B = SAS^{-1}.}$$

(2) Be careful:

$$A \sim B \Rightarrow \sigma(A) = \sigma(B)$$

but $\sigma(A) = \sigma(B) \not\Rightarrow A \sim B$

(If $\sigma(A) = \sigma(B)$, we call A, B <u>co-spectral</u>)

<u>Counterexample</u> (to show cospectral $\not\Rightarrow$ similar)

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\sigma(A) = \{0, 0\} = \sigma(B)$$

But $A \not\sim B$ because if they were similar, then there would exist invertible S such that

$$B = SAS^{-1} = S\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}S^{-1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

<u>contradiction.</u>

<u>Theorem</u> If $A \sim B$, then $\operatorname{rank} A = \operatorname{rank} B$.

$$\left(\begin{aligned} \operatorname{rank} A &= \dim R(A) \\ &= \dim \text{span of columns of } A \end{aligned} \right)$$

<u>Proof</u> Recall "Dimension Theorem":

If $A \in M_n$, then $\operatorname{rank} A + \underbrace{\dim N(A)}_{\text{dimension of nullspace}} = n$

$$\left(N(A) = \operatorname{Null}(A) = \{ x : Ax = 0 \} \right)$$

So to prove our theorem, it is enough to show that if $A \sim B$, then

"$\dim N(A) = \dim N(B)$"

So let $\{ x_1, x_2, \ldots, x_k \}$ be a basis for $N(B)$.

$$(\dim N(B) = k)$$

<u>Claim</u> $W = \{ S^{-1}x_1, S^{-1}x_2, \ldots, S^{-1}x_k \}$ $(B = SAS^{-1})$

is a basis for $N(A)$.

Prove now that $W \subseteq N(A)$ & spanning set of $N(A)$ & linearly independent. (Exercise!) ∎